Vaibhav Dhaka

ML/DL Engineer

+91-6375078486 vaibhavdhaka79@gmail.com linkedin.com/in/vaibhavdhaka

Summary

ML and NLP practitioner experienced in building predictive models using Python and SQL, with a focus on real-world applications.

Skills

Programming: Python, SQL, R, C, C++
ML/DL: Classification, Regression, ANN, CNN
Analytical Abilities: Exploratory Data Analysis (EDA), Insight Generation
Core Concepts: Data Structures, DBMS, OOP

Relevant Experience

Email Spam Classification System — **February 2026 – March 2026**
Self-initiated Project — *Tools Used: Python [Pandas, Scikit-learn]*

- Built an NLP-based spam detection system using Multinomial Naive Bayes on a dataset of **27,000+ emails**.
- Designed a text preprocessing pipeline (tokenization, stemming, stopword removal) to reduce noise and improve model input quality.
- Transformed textual data using TF-IDF, enabling effective numerical representation of high-dimensional text features.
- Trained and evaluated multiple models, selecting Multinomial Naive Bayes based on superior performance, achieving **97.5% accuracy**.
- Evaluated model performance on test data using precision, recall, and confusion matrix, achieving **99% precision** and reducing false positives.

Projects

Movie Recommendation System — **November 2025 – December 2025**
Machine Learning Project — *Tools Used: Python [Pandas, NumPy, nltk]*

- Developed a content-based movie recommendation system using **NLP techniques** on movie metadata.
- Performed text pre-processing including: tokenization, stopword removal, and stemming to clean and normalized data.
- Transformed textual features into numerical vectors using TF-IDF and computed similarity using cosine similarity.
- Designed a recommendation engine to suggest top-N similar movies based on user input.
- Evaluated recommendation quality using **Precision@5 (0.84 relaxed, 0.32 strict)** and **Diversity (0.87)**, and **Average Similarity (0.21)**, balancing relevance and variety.

Customer Churn Analysis — **August 2025 – September 2025**
Deep Learning Project — *Tools Used: Python [Pandas, NumPy, TensorFlow]*

- **Analyzed behavior of 10,000 telecom customers** to identify churn patterns based on tenure, contract type, and monthly charges.
- Engineered new features including contract type buckets and tenure bands to improve segmentation and insights.
- Generated 11 visualizations (histograms, box plots, heatmaps) to highlight churn-prone segments.
- Revealed that 28% of churned users were on monthly contracts with less than 3-month tenure, suggesting the need for early engagement strategies.

Awards & Certifications

- **Energython Hackathon Participant** Developed a team-based solution to optimize waste collection using shortest path algorithms, improving route efficiency and reducing travel time for garbage collection systems.
- **Google Developers Machine Learning Crash Course** Completed hands-on training covering regression, classification, loss functions, and model evaluation, applying data preprocessing and feature engineering techniques on real-world datasets.

Education

B.Tech. Computer Science and Business Systems — **Graduated: 2029**
Vellore Institute of Technology, Vellore — *CGPA: 8.72 / 10*